Homework 2: Apriori

Deadline: Friday 20th February 2026

Overview:

The objective of this homework is to perform **Market Basket Analysis** using the **Apriori algorithm** and extract association rules from transactional data.

Dataset:

You are given a dataset titled ***online_retail.csv.***

Tasks

1. Transaction Construction

- Construct transactions such that each invoice corresponds to one basket.
- Represent each basket as a list of purchased items.

2. Time Based Baskets

- Extract hour from **InvoiceDate**
- Divide transactions into 4 groups: **Morning, Afternoon, Evening, Night**
- Create 4 separate basket collections based on these time groups.

3. One Hot Encoding

- Write your **own function** that converts a list of baskets into a one hot encoded Dataframe.
- Use a **library-based encoder** to do the same.
- Compare both encodings (dimensions and example rows).

4. Frequent Itemset Mining

- Apply the **Apriori algorithm** one each time-based basket.
- Use at least 2 different support thresholds.
- Report:
    - Number of frequent itemsets.
    - Largest itemset length.

5. Association Rule Mining

- Extract rules using **Confidence** and **Lift.**
- Filter rules with **Confidence >= 0.6** and **Lift > 1**

6. Comparative Analysis

Answer the following:

- Which time period produces the strongest rules?
- Which item pair appears in multiple time periods?
- How does changing support affect the number of rules?

Restrictions:

- Do NOT use FP-Growth or MMIS
- Do NOT hardcode transactions
- Do NOT skip Apriori steps
- Only Apriori + association rules are allowed

Starter Code

```
import pandas as pd

from mlxtend.frequent_patterns import apriori, association_rules


df = pd.read_csv("online_retail.csv")


# TODO: create baskets grouped by invoice

baskets = ...


# TODO: split baskets into time groups

morning_baskets = ...

afternoon_baskets = ...

evening_baskets = ...

night_baskets = ...
```

Custom Encoder (Skeleton Only)

```
def encode_transactions(transactions):

    # find all unique items

    items = ...


    encoded = []

    for t in transactions:

        row = {}

        for item in items:

            # encode presence/absence

            ...

        encoded.append(row)


    return pd.DataFrame(encoded)
```

Apriori & Rules

```
encoded_df = encode_transactions(morning_baskets)

freq_items = apriori(encoded_df, min_support=..., use_colnames=True)

rules = association_rules(freq_items, metric="...", min_threshold=...)

rules = rules.sort_values(by="...", ascending=False)
```

Guidelines:

The task should be completed individually.

Submit your assignment with your roll number and homework number. (i231234_HW2)

The assignment must be implemented in **a single Jupyter Notebook**, clearly documented using **Markdown cells and code comments**.

Goodluck 🙂

In case of issues email me at i222036 Umema Ashar